

NAME: _____

PERIOD: _____

Sorting Trays After New Labels

AP Statistics · Unit 3 · Topic 3.7

[STAR] THE PROBLEM

A cafeteria served 5,000 September trays after new sorting-bin labels were installed. The director claims at least 60% are sorted correctly. An SRS of 250 trays finds 136 correct; the club tests at $\alpha = 0.05$.

Imani: “The p -value is 0.0354, so reject H_0 . The labels caused sorting below 60 percent for every month.”

Drew: “The 5.6-point gap may be too small to matter, so it is not significant. Accept the director’s claim.”

The test conditions have been checked: random sample; $250 < 500$; null-expected counts 150 and 100.

September study

Source	Trays	Correct	α
September	5,000	claim: $\geq 60\%$	–
SRS	250	136	0.05

FIRST TAKE — one sentence before you work the parts

Can these September trays tell us what caused the result in every month? Give one reason from the study.

[BOOK] MAKE THE DECISION, THEN BOUND THE CLAIM (keep on the page)

For a one-proportion test, $z = (\hat{p} - p_0) / \sqrt{p_0(1 - p_0)/n}$; for $H_a : p < p_0$, use the left normal tail. If the p -value is at most the predetermined α , reject H_0 ; otherwise fail to reject it. State evidence for H_a in context. Failing to reject does not establish that H_0 is true. Statistical significance uses the evidence rule; practical importance asks whether an effect is large enough to matter. A random sample supports inference to its population. A cause requires a comparison with random assignment. Printed normal table: $P(Z \leq -1.8074) = 0.035351$. The left area is the lower-tail probability.

Work (a)–(c).

DOK 1 (a) Define p and state the hypotheses for checking whether correct sorting is below 60%.

DOK 2 (b) Calculate $\hat{p}$, the one-proportion z statistic, and the lower-tail p -value. Use the verified conditions above.

DOK 3 (c) In 4–5 sentences, correct both reports. Compare the p -value with α , separate statistical significance from practical importance, and limit the conclusion to the people or items studied and what the design can show about cause.

Frame: Because p -value = _____ is _____ α , I would _____ H_0 and conclude _____.

Frame: The result is statistically _____, but the design does not show _____; the 5.6-point gap's practical meaning depends on _____.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored E / P / I.